VF Asia Footwear Laboratory Testing Manual

Test Method for Oil Content	Test Method #: VF-I01 REF #: TM-R- 026 Page 1 of 5
Date Issued: December 12, 2018 Revision #: 2 Reason for revision: Rectify typo error of calculation for % oil content	Created by: Nila Babia Erazo

PURPOSE AND SCOPE: To determine the amount of soluble lubricant or oil in all types of vegetable and chrome tanned leather used in VF footwear by extracting with chloroform.

REFERENCES: ASTM D 3495, SATRA TM 346

EQUIPMENT AND MATERIAL:

1. All glass Soxhlet extraction apparatus, used with 250ml extraction flasks, extraction tube, and condenser
2. Electric heating mantle with 3 burners, capable of holding the 250ml extraction flasks.
3. Cellulose fat –free extraction thimbles, single thickness (33mm x 80 mm)
4. Analytical balance, capable of weighing to 0.001 grams.
5. Forced air circulating oven, capable of maintaining a temperature of 100 +/- 2°C.
6. Desiccator, with suitable desiccant or drying agent such as calcium chloride, calcium sulfate, or silica gel.
7. Solvent , CX1054 OmniSolv Chloroform
8. Absorbent fat-free cotton
9. Scissors
10. Dry leather

SAMPLE PREPARATION:

1. Condition the sample for at least 24 hours at standard laboratory environment at 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.
2. Cut several pieces of leather specimen into approximately 3.18 mm x 3.18 mm.

REPETITIONS: Test two (2) samples per material**.**

TEST PROCEDURE:

Note: Chloroform is a hazardous / flammable material which should be handled carefully. Contact with the skin and inhalation must be avoided. Chloroform should always be used in a working exhaust hood, and protective gloves and safety glasses should be worn.

1. Switch the fume hood ON.

2. Plug the hot plate cord into the electric receptacle

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3. Weigh out the diced leather specimen at approximately 5.800 grams. Record the actual weight **: W_1**
4. Write the material identification, with pencil (ONLY), on the extraction thimble and place diced leather specimen into the thimble.
5. Place the extraction thimble into the extraction tube.
6. Remove an empty dry 250ml boiling flask from the front of the desiccator (assures flask has been stored at room at room temperature (20°C) for 30 minutes minimum). Write the material identification on the boiling flask and weigh it on the gram scale to the nearest 0.001g. Record the weight of the flask.: **W_2**
7. Measure out 150ml of chloroform and pour into 250ml boiling flask. Wipe any spilled chloroform on the boiling flask with absorbent towel.
8. Turn on water valve and open hose release clamp to allow water to fill condenser. Allow water to trickle from hose into drain throughout entire test.
9. Place the partially filled 250ml boiling flask on the hot plate testing station.
10. Loosen the wing nut holding the condenser and insert into the extraction tube. Gently lower the condenser and extraction tube into the boiling flask. Finger tighten the wing nut to secure the assembly in place.
11. Turn the testing station hot plate ON.
12. Set the timer for 6 hours (40 to 50 flushes at an approximate 7 - 8 minutes cycle time). If bubbling/foaming is observed in the extraction tube and/or the boiling flask, check to be sure the connection of the condenser, extraction tube and the boiling flask is sealed properly so that air is not introduced into the process. Reseal if required. If the leather has excess emulsifiers (surface-active agents such as soap), a foaming action usually takes place in the extraction tube and the boiling flask during the extraction process. If this happens, ignore it, as it does not affect the overall test results. See figure 1 for comparison test method and flush cycles.
13. At the end of the test, and after all solvent has flushed into the boiling flask, loosen the wing nut holding the condenser and gently separate the extraction tube and the boiling flask from the condenser. Reposition the condenser and secure in place by tightening the wing nut.
14. Gently separate the boiling flask from the extraction tube and set it aside.

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15. Pour any remaining solvent from the extraction tube into the boiling flask. Block the open end of the extraction tube to prevent the extraction thimble from slipping out. Remove extraction thimble from extraction tube.

16. Replace boiling flask on the hot plate. Loosen the wing nut holding the condenser and insert it into the extraction tube and gently lower the assembly into the boiling flask. Tighten the wing nut to secure the assembly in place.

17. Allow the solvent to reduce down to a point where the solvent just covers the bottom of the boiling flask - approximately 1½" dia. (It's important not to burn off residuals). As the solvent reduces down, the chloroform will distill into the extraction tube. Do not allow the chloroform to 'flush'. Prior to 'flushing', go to step **18**. If there is still an excessive amount of solvent left in the boiling flask then continue back at step **16.**

    NOTE: This is a critical step. If all liquid evaporates, then test will need to be discontinued and started over.

18. Distill (reclaim chloroform):

- *Before the distilled chloroform flushes into the boiling flask, loosen the wing nut holding the condenser and gently separate the extraction tube and the boiling flask from the condenser. Reposition the condenser and secure in place by tightening the wing nut. Turn the hot plate OFF. Shut water valve OFF.*

- *If the distilled chloroform in the extraction tube is CLEAR, pour it into the jar labeled "Distilled Chloroform". Separate the boiling flask from the extraction tube and set aside.*

- *If the distilled chloroform in the extraction tube is discolored (NOT CLEAR), pour it into a clean boiling flask, place the flask on the hot plate burner, and set the temperature dial to #3. Reduce the chloroform down until it completely evaporates. (**Do not allow the discolored chloroform to bubble or foam up over the top of the flask and drip into the hot plate burner, as this can result in damage to the burner heating element. If necessary, reduce the temperature setting to a point that does not allow the bubbling or foaming to rise up over the top of the boiling flask).***

19. Place the boiling flask with the solvent extraction that has been reduced in a 100°C (212°F) pre-heated oven for approximately 15 - 16 hours (overnight). Remove from the oven and cool to room temperature in the desiccator for approximately two (2) hours.

20. Weigh boiling flask with specimen on gram scale to the nearest 0.001g. Record the weight**.** **W_3**

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CLEAN UP of APPARATUS:

- Pour just enough chloroform into the used boiling flask to cover the bottom and soak for about 1 hour. Then pour used chloroform into glass beaker located under the fume hood.
- Pour ¼ teaspoon of Alconox detergent into the used boiling flask and fill halfway with warm water soak for 1 hour. Clean with a small bottlebrush. Empty and rinse thoroughly with plain water. Drain and place clean boiling flask in 100°C (212°F) oven to dry. Store clean boiling flask in desiccator after completely dry. Place flasks just cleaned to back of desiccator and rotate previously cleaned flasks to front of the desiccator.

CALCULATION:

Percent oil content (%) = $[(W_3 - W_2) / W_1)] \times 100$

Where:

W_1 = original weight of extracted leather

W_2 = original weight of 250ml boiling flask

W_3 = weight of 250ml boiling flask and matter

REPORT:

1. Percent oil content (%)
2. Report material description, and any deviations from this procedure.

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Figure 1. Oil Extraction Method/Cycle Comparison Table

Method	#Hours	Cycle Times	# Flushes	Chemical	Approx. Solvent Volume
Footwear Competitor	6 hours	15 min ± 3 min	24 times	Methylene Chloride	150ml
ASTM (ACLA B4) Tannery	4 hours	12 min	20 times	Hexane	150ml
ACLA B4 Tannery	4 hours	15 min ± 3 min	16 times	Chloroform	150ml
ACLA B4 Tannery	2 hours	2 min	60 times	Chloroform	90ml
ACLA B4 Tannery	5 hours	15 min ± 3 min	20 times	Petroleum Ether	150ml
Adhesive Supplier	6 hours	n/a	n/a	Methylene Chloride	90ml
Fed Test Method 311	2 hours	2 - 3 min	60 times	Chloroform	90ml
ASTM D-3495-83	5 hours	6 min	50 times	Hexane	150ml
Satra CM46	4 hours	6 min	40 times	Dichloromethane	150ml